

Formulation workflows: from screen to handoff

Julia Carver, Formulation Development Lead | 09/18/2026 | v3 FINAL final | Durnesswick Laboratories Inc, a Tarnovell company

Make the next preparation decision

- Choose a formulation direction that matches intended use.
- Expose handling limits while the formulation set is still exploratory.
- Keep the reason for selection with the recipe card.

Where we work in early CMC

- Vehicle ladder and Excipient compatibility screen work packages.
- Follow-up on dilution, settling, recovery and preparation order.
- Pilot-lot preparation and practical transfer discussions.

A decision-led workflow

Step	Output
Define intended use	Agreed dose and preparation question
Allocate material	Screen and follow-up allowance
Compare formulation set	Traceable bench-sheet observations
Resolve failure modes	Selection rationale
Handoff at scale gate	Recipe card and receiving criteria

Intake: material and context

- Ashley King coordinates receipt and allocation.
- Supply identity, physical form, material-lot relationship and available mass.
- Tell us whether the same material condition will be used in the next preparation.

Intended use sets the comparison

- Dose and administration volume determine the loading question.
- Dilution medium and hold conditions determine relevant stress.
- Exposure use and product storage require different evidence.

Screen concentration in mg/mL

- State whether concentration refers to dissolved material or total dispersion loading.
- Keep the target distinct from measured equilibrium capacity.
- Do not carry a nominal concentration into a solution claim after precipitation.

Loading and excipient burden

- Use weight percent when describing excipient loading.
- Solvent support can change apparent preparation feasibility.
- Viscosity may limit mixing and withdrawal.
- Dilution can reverse an apparent concentrate advantage.

Vehicle ladder preparation

- Define recipe composition and addition order.
- Capture wetting, mixing and initial appearance on the bench sheet.
- Preserve unsuccessful preparations so later changes have context.

Vehicle ladder observation

- Compare initial, standing and post-dilution behavior.
- Record whether sediment recovers under the intended mixing step.
- Interpret capacity beside handling rather than choosing on a solubility ranking alone.

Dilution follow-up

- Use the medium relevant to the intended administration step.
- Keep the concentrate recipe linked to the diluted observation.
- A failure requires a revised vehicle or dosing approach, followed by the relevant comparison.

Excipient compatibility screen design

- Agree candidate excipients and material condition.
- Choose observation conditions that answer the development question.
- Separate compatibility screening from a product stability commitment.

Screen suitability before recommendation

Question	Required clarity
Which material?	Identity, lot and physical form
Which recipe?	Composition and preparation order
Which condition?	Loading, hold and dilution context
Which exception?	Visible failure retained in the interpretation

Particle-size decisions in μm

- Use measured distributions when numerical size limits are required.
- Consider wetting, sediment strength and redispersion beside size.
- A finer material is not automatically easier to dose or manufacture.

Manufacturability limits

- Mixing and addition sequence may be volume-dependent.
- Container geometry can affect sediment recovery.
- Material loss can change practical preparation yield.
- A small exploratory preparation does not establish a reproducible larger process.

Exploratory output: what you receive

- Bench sheets with recipe identity and observations.
- Development note explaining selection and rejected routes.
- Recipe card with preparation and handling instructions for the defined use.

Pilot-lot scale gate

- Agree preparation volume, mixing method and hold condition.
- Identify differences from the exploratory preparation before execution.
- Carry exceptions forward so the receiving team can recognize a changed outcome.

Receiving-team handoff criteria — revised v3

Criterion	Discussion required
Recipe understood	Composition and sequence are reproducible locally
Material comparable	Lot and physical form match the intended use
Handling workable	Resuspension and withdrawal can be performed
Stop condition explicit	Failure to recover or changed appearance triggers escalation

Scientific and transfer ownership

- Julia Carver: development interpretation and recommendation.
- Miguel Serrano: formulation-set preparation and supporting observations.
- Lena Patel: technical transfer and receiving-team criteria.
- Receiving team: local preparation constraints and intended application.

Commitments at the next scale gate

- Screening concentration, loading and handling claims apply to the examined conditions.
- Process lock, storage claims and dose uniformity require their own evidence.
- Share the next preparation question with Julia Carver; agree the work package before expanding use.